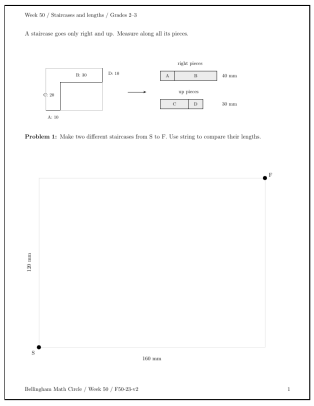
Week 50 / Staircases and lengths / Problem menu

for adults: pick, then print


Week 50 / Staircases and lengths / Grades 2–3

A staircase goes only right and up. Measure along all its pieces.

right pieces

D: 30 D: 10 A B 40 mm

C: 20 up pieces

C D 30 mm

A: 10

Problem 1: Make two different staircases from S to F. Use string to compare their lengths.

F

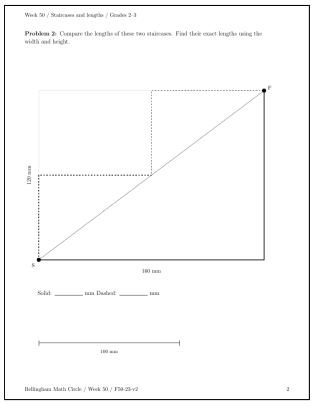
120 mm

S

160 mm

Bellingham Math Circle / Week 50 / F50-23-v2 1

Two different staircases
K–1 P1 · Grades 2–3 P1 · Grades 4–5 P1
first route


Week 50 / Staircases and lengths / Grades 2–3

Problem 2: Compare the lengths of these two staircases. Find their exact lengths using the width and height.

F

120 mm

S

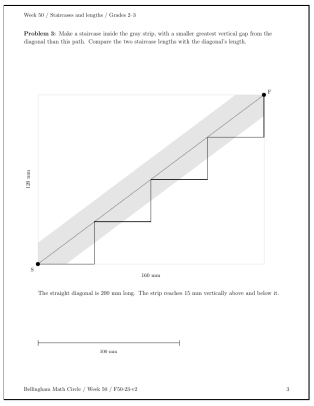
160 mm

Solid: _____ mm Dashed: _____ mm

100 mm

Bellingham Math Circle / Week 50 / F50-23-v2 2

A one-step and a two-step staircase
K–1 P2 · Grades 2–3 P2 · Grades 4–5 P2
first route
wording differs by band


Week 50 / Staircases and lengths / Grades 2–3

Problem 3: Make a staircase inside the gray strip, with a smaller greatest vertical gap from the diagonal than this path. Compare the two staircase lengths with the diagonal's length.

F

120 mm

S

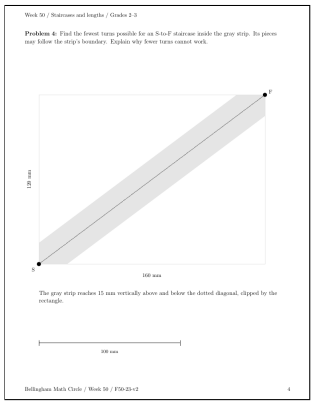
160 mm

The straight diagonal is 200 mm long. The strip reaches 15 mm vertically above and below it.

100 mm

Bellingham Math Circle / Week 50 / F50-23-v2 3

A staircase closer to the diagonal
K–1 P3 · Grades 2–3 P3 · Grades 4–5 P3
first route
wording differs by band


Week 50 / Staircases and lengths / Grades 2–3

Problem 4: Find the fewest turns possible for an S-to-F staircase inside the gray strip. Its pieces may follow the strip's boundary. Explain why fewer turns cannot work.

F

120 mm

S

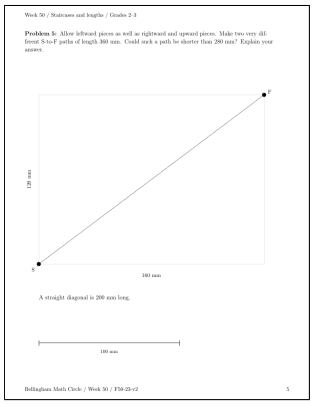
160 mm

The gray strip reaches 15 mm vertically above and below the dotted diagonal, clipped by the rectangle.

100 mm

Bellingham Math Circle / Week 50 / F50-23-v2 4

Fewest turns inside the strip
K–1 P4 · Grades 2–3 P4 · Grades 4–5 P4
return route
wording differs by band


Week 50 / Staircases and lengths / Grades 2–3

Problem 5: Allow leftward pieces as well as rightward and upward pieces. Make two very different S-to-F paths of length 360 mm. Could such a path be shorter than 280 mm? Explain your answer.

F

120 mm

S

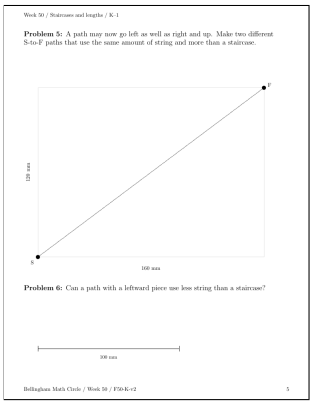
160 mm

A straight diagonal is 200 mm long.

100 mm

Bellingham Math Circle / Week 50 / F50-23-v2 5

Paths that may go left
K–1 P5 · Grades 2–3 P5
return route
wording differs by band


Week 50 / Staircases and lengths / K–1

Problem 5: A path may now go left as well as right and up. Make two different S-to-F paths that use the same amount of string and more than a staircase.

F

120 mm

S

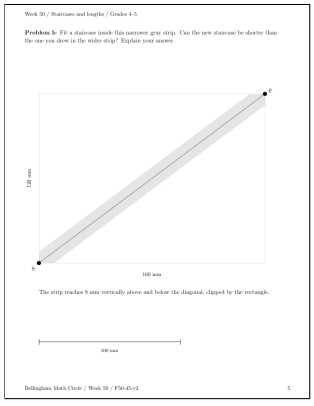
160 mm

Problem 6: Can a path with a leftward piece use less string than a staircase?

100 mm

Bellingham Math Circle / Week 50 / F50-K-v2 5

A leftward piece and less string
K–1 P6
return route


Week 50 / Staircases and lengths / Grades 4–5

Problem 8: Fit a staircase inside this narrower gray strip. Can the new staircase be shorter than the one you drew in the wider strip? Explain your answer.

F

120 mm

S

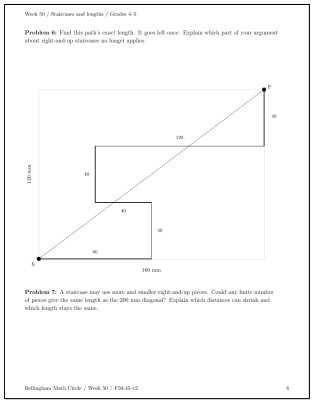
160 mm

The strip reaches 8 mm vertically above and below the diagonal, clipped by the rectangle.

100 mm

Bellingham Math Circle / Week 50 / F50-45-v2 5

A narrower strip
Grades 4–5 P5
return route


Week 50 / Staircases and lengths / Grades 4–5

Problem 6: Find this path's exact length. It goes left once. Explain which part of your argument about right-and-up staircases no longer applies.

F

120 mm

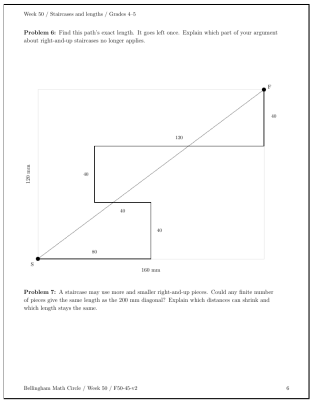
S

160 mm

Problem 7: A staircase may use more and smaller right-and-up pieces. Could any finite number of pieces give the same length as the 200 mm diagonal? Explain which distances can shrink and which length stays the same.

Bellingham Math Circle / Week 50 / F50-45-v2 6

The exact length of a path that goes left once
Grades 4–5 P6
return route


Week 50 / Staircases and lengths / Grades 4–5

Problem 6: Find this path's exact length. It goes left once. Explain which part of your argument about right-and-up staircases no longer applies.

F

120 mm

S

160 mm

Problem 7: A staircase may use more and smaller right-and-up pieces. Could any finite number of pieces give the same length as the 200 mm diagonal? Explain which distances can shrink and which length stays the same.

Bellingham Math Circle / Week 50 / F50-45-v2 6

Could more, smaller pieces reach 200 mm?
Grades 4–5 P7
return route